

Zichong Li

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Research Interests

My research focuses on efficient and reliable large language model (LLM) training. I study methods that serve this goal across the training pipeline, including optimizer and architecture design for pretraining, as well as training algorithms for mid- and post-training.

Education

Georgia Institute of Technology

Ph.D. in Machine Learning

Advisor: Prof. Tuo Zhao

Atlanta, GA

Aug. 2023 – Present

University of Science and Technology of China

M.S. in Data Science; Rank: 1/56

Hefei, China

Sept. 2020 – June 2023

University of Science and Technology of China

B.S. in Mathematics and Applied Mathematics / Probability Statistics

Hefei, China

Sept. 2016 – June 2020

Professional Experience

Applied Deep Learning Research Intern, NVIDIA

May 2026 – Present

- Topic: Token-Efficient Reinforcement Learning for LLM Reasoning
- Mentor: Wei Ping

Research Intern, Microsoft

June 2025 – Aug. 2025

- Topic: Long-Context Adaptation of LLM
- Mentor: Chen Liang.

Research Intern, Microsoft

May 2024 – Aug. 2024

- Topic: Structured Compression and distillation for Large MoE Models
- Mentor: Chen Liang.

Research Experience

Efficient LLM Pretraining. Developed scalable optimizers for compute- and memory-efficient LLM pretraining.

Highlights: NorMuon set new records on the modded-nanogpt benchmark and was integrated into nanochat project.

Structured Compression and Distillation. Developed structured pruning and knowledge distillation strategies for deployable and compact LLMs.

Highlights: SlimMoE released Phi-mini-MoE and Phi-tiny-MoE, compact MoE instruction models with 850K+ total downloads.

RL Post-Training and Long context. Developed RLHF methods for reliable preference optimization and robust reward learning, as well as training algorithm for long context adaptation.

Other Topics. Developed neural point process models, uncertainty quantification methods, efficient ensemble inference systems, and sparse learning software.

Selected Publications

Language Models

- **NorMuon: Making Muon More Efficient and Scalable**
Zichong Li*, Liming Liu*, Chen Liang, Weizhu Chen, Tuo Zhao
ICML 2026. Spotlight, top 2.2%
- **Shuffle the Context: RoPE-Perturbed Self-Distillation for Long-Context Adaptation**
Zichong Li, Chen Liang, Liliang Ren, Tuo Zhao, Yelong Shen, Weizhu Chen
ICML 2026
- **COSMOS: A Hybrid Adaptive Optimizer for Efficient Training of Large Language Models**
Liming Liu, Zhenghao Xu, Zixuan Zhang, Hao Kang, Zichong Li, Chen Liang, Weizhu Chen, Tuo Zhao
ICLR 2026
- **SlimMoE: Structured Compression of Large MoE Models via Expert Slimming and Distillation**
Zichong Li, Chen Liang, Zixuan Zhang, Ilgee Hong, Young Jin Kim, Weizhu Chen, Tuo Zhao
COLM 2025

- **LLMs Can Generate a Better Answer by Aggregating Their Own Responses**
Zichong Li, Xinyu Feng, Yuheng Cai, Zixuan Zhang, Tianyi Liu, Chen Liang, Weizhu Chen, Haoyu Wang, Tuo Zhao
arXiv 2025
- **Mitigating Tail Latency for On-Device Inference with Load-Balanced Heterogeneous Models**
Mu Yuan, Lan Zhang, Di Duan, Liekang Zeng, Miao-Hui Song, **Zichong Li**, Guoliang Xing, Xiang-Yang Li
IEEE TMC 2025
- **Adaptive Preference Scaling for Reinforcement Learning with Human Feedback**
Ilgee Hong*, **Zichong Li***, Alexander Bukharin, Yixiao Li, Haoming Jiang, Tianbao Yang, Tuo Zhao
NeurIPS 2024. *Equal contribution
- **Robust Reinforcement Learning from Corrupted Human Feedback**
Alexander Bukharin, Ilgee Hong, Haoming Jiang, **Zichong Li**, Qingru Zhang, Zixuan Zhang, Tuo Zhao
NeurIPS 2024

Point Processes, and Others

- **Beyond Point Prediction: Score Matching-based Pseudolikelihood Estimation of Neural Marked Spatio-Temporal Point Process**
Zichong Li, Qunzhi Xu, Zhenghao Xu, Yajun Mei, Tuo Zhao, Hongyuan Zha
ICML 2024
- **SMURF-THP: Score Matching-based Uncertainty Quantification for Transformer Hawkes Process**
Zichong Li, Yanbo Xu, Simiao Zuo, Haoming Jiang, Chao Zhang, Tuo Zhao, Hongyuan Zha
ICML 2023
- **Efficient Deep Ensemble Inference via Query Difficulty-dependent Task Scheduling**
Zichong Li, Lan Zhang, Mu Yuan, Miaohui Song, Qi Song
ICDE 2023
- **CoTel: Ontology-Neural Co-Enhanced Text Labeling**
Miaohui Song, Lan Zhang, Mu Yuan, **Zichong Li**, Qi Song, Yijun Liu, Guidong Zheng
WWW 2023
- **Transformer Hawkes Process**
Simiao Zuo, Haoming Jiang, **Zichong Li**, Tuo Zhao, Hongyuan Zha
ICML 2020
- **PRIMAL: A Linear Programming-based Sparse Learning Library in R and Python**
Qianli Shen*, **Zichong Li***, Yujia Xie, Tuo Zhao
Software. *Equal contribution

Technical Skills

Programming: Python, C, R, LaTeX

Machine Learning: Large Language Models, Transformers, long-context modeling, mixture-of-experts models, RLHF, reinforcement learning, model compression, distillation, efficient optimization, uncertainty quantification

Tools: PyTorch, Hugging Face, Linux, Git, MATLAB, Mathematica

Honors & Awards

2019 – Outstanding Student, University of Science and Technology of China

2017 – Endeavour Scholarship, University of Science and Technology of China

2016 – First-level Freshman Scholarship, University of Science and Technology of China